Chapter 3 :

Combinatorial Analysis

The aim of combinatorial analysis is to count the diffrent arrangements that can be formed from a set of elements. It is a branch of Mathematics that studies how to count objects. It consists of a set of formulas which are used in several elementary problems in probability theory.

Example 1 : License plates of certain country contain three letters followed by three numbers. How many different plates can we have ?

3. 1. Basics

Definition 3. 1. 1 (Factorial)

The factorial of positive integer number *n* is the product of all positive integers less than or equal to *n*

$$n! = 1 \times 2 \times ... \times (n-1) \times n$$

By convention, the factorial of 0 is equal to 1, i.e. 0 !=1

Definition 3. 1. 2 (Set)

A set is a collection of objects called elements

Definition 3. 1. 3 (Subset)

F is a subset of (or included in) the set *E*, if all the elements of *F* are also the elements of *E* and we note : $F \subset E$

Definition 3. 1. 4 (Cardinal)

The number of elements of a set *E* is called **cardinal of *E***, denoted Card(*E*) or $|E|$

Note 1 :

We denoted by ϕ the empty set (or null set). It is the set that does not contain any element.

Example 2. $A = \{1, 2, 3, a, b, c\}$ $B = \{1, 2, 3\}$.

We have : $B \subset A$, $|A| = 6$, and $|B| = 3$

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3. 1. 1 Fundamental counting principle

The fundamental principle of counting make it possible to count the number of experimental results which can be decomposed into a succession of sub-experiments.

Property 1 (The case of two experiments). Suppose that two experiments need to be carriend out. If experiment 1 can produce any one of m results and if, for each of them, there are n possible results for experiment 2, then there are $m \times n$ results for the two experiments taken together.

When ther are more than two experiments to be performed, the fundamental counting principle can be generalized as follows :

Property 2 (The case of *m* experiments). Suppose that an experiment is the succession of *m* sub-experiments. If the ith experiment has n_i possible outcomes, then the total number *n* of possible outcomes of the overall experiment is :

$$n = \prod_{i=1}^{m} n_i = n_1 \times n_2 \times ... \times n_m$$

Example 3. A padlock has 3 numbers from 0 to 9. We therefore fave three choices to make successively :

- First digit of the code (the first sub-experiment) : we have $n_1 = 10$ possibilities.
- Second digit of the code (the second sub-experiment) : we have $n_2 = 10$ possibilities.
- Third digit of the code (the third sub-experiment) : we have $n_2 = 10$ possibilities.

The total number of possibilities *n* of different codes that can be constructed is the product of the possibilities of each sub- experiment :

$$n = \prod_{i=1}^{3} n_i = n_1 \times n_2 \times n_3 = 1000$$

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3. 2. Counting formula

3. 2. 1 Arrangement

Definition 3. 2. 1. Let *E* be a finit set with *n* elements and *p* an integer satisfying $0 \leq p \leq n$. We call arrangements of *p* elements, all ordred sequences of *p* elements taken from the *n* elements.

We distinguish two cases :

1) **Arrangement with repetition :** When an element can be observed several times in an arrangement, the number of arrangement with repetition of *p* elements among *n*, is given by :

$$\tilde{A}_n^p = n^p$$

2) **Arrangement without repetition :** When each element can only be observed once in an arrangement, the number of arrangements without repetition of *p* elements among *n* is :

$$A_n^p = \frac{n!}{(n-p)!}$$

Example 4. We want to create a password composed of 3 digits. Here, the order of elements is important (for example, password 112 is different from password 211). In addition, we can have a digit that repeats. Therefor, the number of possible cases corresponds to the number of arrangements with repetition, with $p = 3$ and $n = 10$:

$$\tilde{A}_{10}^3 = 10^3$$

On the other hand, if we requiere that the password must contain 3 different digits, the number of passwords possible corresponds to the the number of arrangements without repetition, with $p = 3$ and $n = 10$:

$$A_{10}^3 = \frac{10!}{(10-3)!} = 8 \times 9 \times 10 = 720$$

3. 2. 2 Permutation

Definition 3. 2. 2. Given a set *E* of *n* elements. We call permutation of n elements any ordered sequence of *n* elements. We distinguish two cases :

1) **Permutation without repetition :** When all elements are distinct, the number of parmutations without repetition of n elements is given by :

$$P_n = n!$$

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2) Permutation with repetition : In the case where there exist several repetitions n_i (with $\sum_{i=1}^{k} n_i = n$) of k elements among the n elements, the number of permutation with repetitions is given by :

$$\tilde{P}_n = \frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

Example 5. How many ways can you arrange 5 distinct books on a shelf ?

This is the case of prmutation without repetition, therefore :

$$P_5 = 5! = 120$$

Now, if ampng the 5 books we have 3 identical mathematics books, and 2 identical computer science books. The number of ways to organize these books is :

$$\tilde{P}_5 = \frac{5!}{3! \times 2!} = 30$$

3. 2. 3 Combination

Definition 3. 2. 3. Given a set *E* of *n* elements, we call combination of *p* elements any subset of *p* elements taken from the *n* elements, without any determined order and without replacement.

The number of combinations of *p* elements taken from *n*, is given by :

$$C_n^p = \frac{n!}{p!\,(n-p)!}$$

Example 6. If a student must respond to 3 questions from 4 on an exam. How many ways can be respond ?

The exam's questions are distincts, and the order in wich the student answers the questions is not important. The number of possible ways is a combination with $p = 3$ and $n = 4$

$$C_4^3 = \frac{4!}{3!\,(4-3)!} = 4$$

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Properties 1

- $C_n^p = C_n^{n-p}$ (symmetry)
- $C_{n-1}^{p-1} + C_{n-1}^p = C_n^p$ (Pascal's formula)
- $C_n^n = C_n^0 = 1$
- $C_n^1 = C_n^{n-1} = n$

	Arrangement	Permutation	Combination
Without repetition	$A_n^p = \frac{n!}{(n-p)!}$	$P_n = n!$	$C_n^p = \frac{n!}{p!\,(n-p)!}$
With repetition	$\tilde{A}_n^p = n^p$	$\tilde{P}_n = \frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$	

Table 3.1 : Counting formulas

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